ZKBlockchain

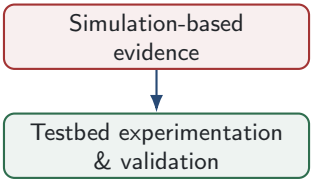
A Privacy-Preserving Blockchain Framework for Healthcare Data Sharing Using Post-Quantum Cryptography and Zero-Knowledge Proofs

Daniyal Qureshi (23I-0645) • Zia Ul Baqa (23I-0606) • Hannan Abid (23I-0713)
Supervisor: Dr. Noshina Tariq

Fall 2026 • FYP-I Proposal Defense

Problem Statement

STREAM: RESEARCH-DERIVED



Existing simulation-based blockchain architectures for Internet hospital systems demonstrate promising performance, but their security, privacy, scalability and reliability claims remain **insufficiently validated** under real cryptographic implementations, distributed network conditions and realistic physiological data.

Stakeholders & Current Data Workflow

Primary Stakeholders

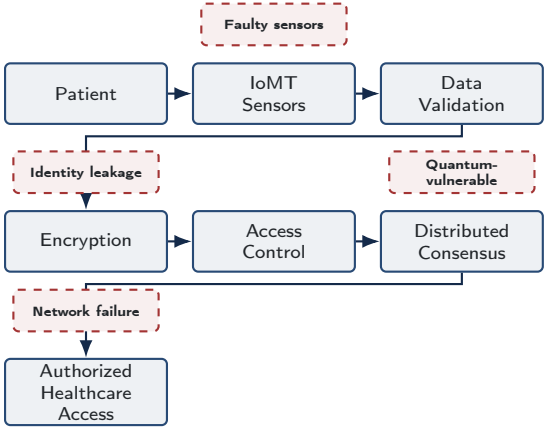
- Doctors / Healthcare professionals – authorized, timely access
- Patients – privacy & data integrity

Secondary Stakeholders

- Hospital / Healthcare Institutions

SDG 3 – Good Health & Well-Being

SDG 9 – Industry, Innovation & Infrastructure



Existing Solution & Evidence-Based Gap

WHAT THE PAPER SHOWS → **WHAT IT DOES NOT YET ESTABLISH** → **WHAT OUR FYP WILL TEST**

Capability	Existing Paper	Gap	Proposed FYP
Post-Quantum Cryptography	Kyber-inspired	No standard PQC	Standardized PQC
Access Privacy	Hash commitment	No true ZK proof	Zero Knowledge Proof
Consensus	Simulated Raft	No real deployment	Multi-node HotStuff
Network	Constant delay	Unrealistic WAN	Delay, loss & failures
Sensor Data	Synthetic	Limited realism	Framingham Dataset
Security	Theoretical claims	Limited validation	Empirical threat testing

*Note: gaps are framed as **validation and implementation limitations**, not as invalidating the original paper's contribution.*

Research & Technical Basis

1. IoMT Data Reliability

- Noise, missing and spoofed readings
- Validate using realistic physiological data

2. Post-Quantum Cryptography

- RSA vulnerable to quantum attacks
- Standardized PQC over ad-hoc simulation
- Candidate: *ML-KEM*

3. Privacy-Preserving Access

- Hash commitment $\neq$ true ZK proof
- Minimize identity/role exposure
- Candidate: *Zero-Knowledge Proofs*

4. Distributed Consensus

- Raft (CFT) vs. BFT alternatives
- Require Byzantine fault tolerance
- Candidate: *HotStuff*

Decision Table

Technical Question	Alternatives	Selected	Reason
PQC primitive?	ML-KEM / HQC	*ML-KEM*	NIST-standardized, quantum-resistant
Privacy mechanism?	RBAC / Hash / ZK	*ZK Proof*	Verification without identity exposure
Consensus?	Raft / PBFT / HotStuff	*HotStuff*	Byzantine resilience at feasible scale

Why This Is a Complex Computing Problem

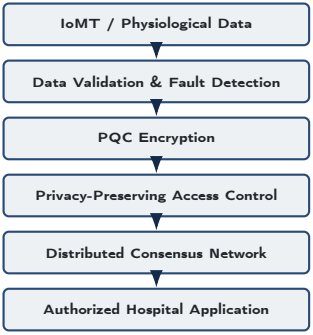
This is not routine CRUD / system-integration work. It requires balancing conflicting engineering trade-offs:

- **[T1] Security ↔ Performance** — stronger crypto/privacy increases computation & latency
- **[T2] Privacy ↔ Verifiability** — prove authorization without exposing identity/policy
- **[T3] Scalability ↔ Fault Tolerance** — more nodes change communication overhead & failure behavior
- **[T4] Reliability ↔ Complexity** — faulty sensor data cannot simply be trusted and stored

Seoul Accord Graduate Attributes Addressed

Complex problem analysis • Design under uncertainty • Trade-off evaluation • Integration of multiple technical domains (cryptography, distributed systems, networking, access control) • Experimentation and empirical validation

Proposed Contribution: System Architecture



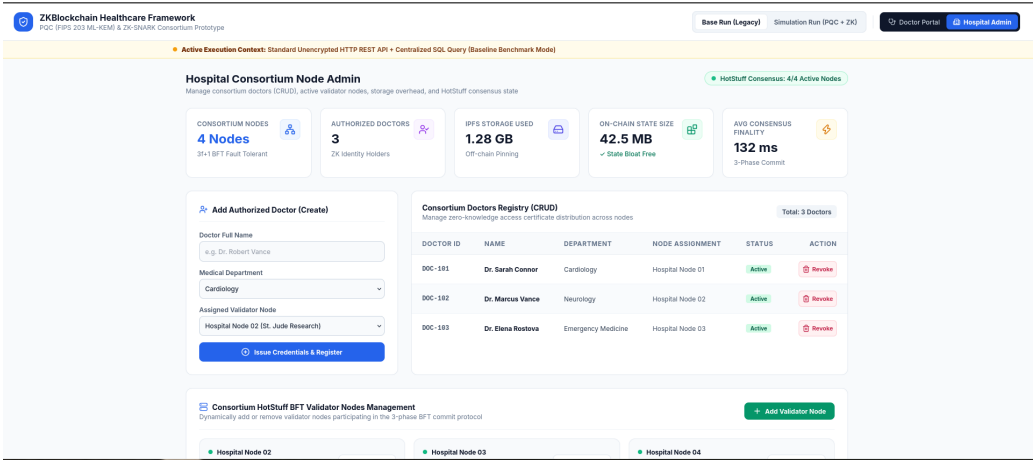
In Scope

- Real PQC implementation
- Privacy-preserving authorization
- Distributed consensus framework
- Realistic network simulation/emulation
- Physiological/IoMT dataset validation
- Measurable benchmarking
- Testbed Implementation

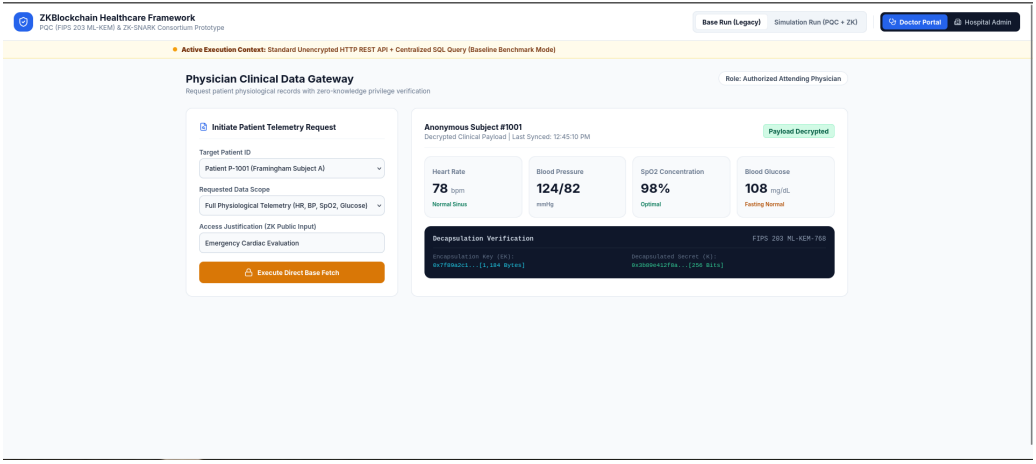
Out of Scope

- Clinical diagnosis
- Production-scale infrastructure
- Regulatory certification claims
- Clinical efficacy claims

Proof of Concept (POC) - Hospital's Dashboard



Proof of Concept (POC) - Doctor's Dashboard



Evaluation Plan

Research Question	Evaluation Metric
Does real PQC give acceptable performance?	Latency, throughput, ciphertext size
Does the privacy mechanism reduce information exposure?	Correctness
Does consensus remain usable under realistic networks?	TPS, latency, recovery
Does sensor correction generalize beyond synthetic data?	Accuracy, MAE, Fault Detection

Risks & Responsible Use of GenAI

Key Project Risks → Mitigation

- **ZK Integration Complexity** → Well-defined proof prototype benchmarked against baseline hash commitments.
- **Dataset Unsuitability** → Secondary open physiological dataset evaluated with identical protocols.
- **WAN Deployment Constraints** → Controlled network emulation across distributed container nodes.
- **PQC Debugging Overhead** → Incremental scaling starting from a minimal cluster configuration.

Responsible Use of GenAI

- **Code Scaffolding:** Limited strictly to boilerplate generation, formatting, and unit test scripts.
- **Human-in-the-Loop:** 100% manual review and security auditing of all AI-suggested code.
- **Data Protection:** Zero credentials, private keys, or sensitive telemetry shared with AI models.
- **Core Security Logic:** Zero reliance on GenAI for cryptography algorithms or consensus rules.
- **Full Transparency:** Complete scope of AI assistance documented in repository and thesis.

Team Responsibilities & FYP-I / FYP-II Plan

Hannan Abid
Post-quantum Cryptography
Implementation & Secure Data Storage

Daniyal Qureshi
Privacy-Preserving / Zero-Knowledge
Access, Control & Security Evaluation

Zia Ul Baqa
Distributed Consensus, Network Realism,
IoMT Data Pipeline & Fault Tolerance

Shared: System Integration • Testing • Benchmarking • Documentation • Security Review

FYP-I

Literature & Gap validation → Technical Spike / POC → Baseline Implementation → Dataset Preparation → Initial Experiments → Gaps Implementation → Feasibility Study

FYP-II

Full integration → Testbed Evaluation → Performance Benchmarking → Final Refinement → Demonstration & Thesis

Ethical Constraints & Legal Considerations

ETHICAL CONSTRAINTS

1. **De-identified Data**
   Use only publicly released research data; no live-patient data.

2. **No Clinical Use**
   Academic prototype only — no diagnosis or clinical decisions.

3. **Defensive Security**
   PQC/ZK used only for privacy and security evaluation.

LEGAL & REGULATORY

4. **Data Protection**
   Follow applicable data-protection and healthcare privacy requirements.

5. **On-Chain Privacy**
   Store proofs/references on-chain — never raw health records.

6. **Law & Licensing**
   Comply with PECA and all dataset/library license usage requirements.